

Member ID: _____

Time: _____

Rank: _____



FUNDAMENTAL SPREADSHEET APPLICATIONS (230)

STATE 2022

Production:

Job 1: Worksheets with answers and formulas	_____ (180 points)
Job 2: Formatting Worksheet	_____ (95 points)
Job 3: Customized Chart	_____ (60 points)
TOTAL POINTS	_____ (335 points)

Test Time: 90 minutes

GENERAL GUIDELINES:

Failure to adhere to any of the following rules will result in disqualification:

1. Member must hand in this test booklet and all printouts if any.
2. No equipment, supplies, or materials other than those specified for this event are allowed in the testing area. No previous BPA tests and/or sample tests (handwritten, photocopied, or keyed) are allowed in the testing area.
3. Electronic devices will be monitored according to ACT standards.
4. Put your Member ID and printout number in the right section of the footer on each printout unless instructed otherwise. Your name or initials should *not* appear on any work you submit.
5. If you finish before the end of the testing time, notify the proctor. Time may be a factor in determining the winner in the event of a tie.
6. When turning in your contest, the jobs should be arranged in printout order.

SCENARIO:

You work for Digital Solution's Employment Consulting Firm. Today your manager asked you to create a spreadsheet and chart comparing last year's January - April Business Positions to this year's Business Positions found. Your company works with multiple cities throughout the United States and Canada.

Job 1 (Worksheets with answers and formulas)	Points Possible	Points Earned
Member ID and Job # in right section of footer	10	
Verdana Size 10 font used for data	10	
Landscape orientation with narrow margins	10	
Worksheet printed landscape on one page centered vertically and horizontally	10	
Typed in cell A1 Digital Solution's Employment Consulting Firm, A2 Business Positions, I1 Previous Year Total, I3 This Year, I5 % Increase, J1 4277	20	
Information typed correctly according to image	20	
In cell F4 formula that calculates the total of the four months by city and fill down	10	
In Cell B13 place a function that calculates each month's positions filled and fill to cell F13	10	
Cell G4, place a formula that calculates the percentage of the total and fill to cell G12.	10	
Inset a function in Cell B15 that averages January's filled positions for the month and fill to cell E15.	10	
Insert a function in cell B16 that calculates the maximum filled positions for the month and fill to cell E16.	10	
Insert a function in cell B17 that calculates the lowest filled positions for the month and fill to cell E17.	10	
Place the cell location for the total filled positions for all cities and months in cell J3.	5	
In cell J5 place a complex formula that calculates the % increase in filled positions from last year to this year.	5	
Insert a column to the right of each month	5	
Title the cell of the added columns to the month % (January %).	5	
Insert a formula to calculate the percentage of each month for each city with the total of that month.	10	
Print a copy of the worksheet with gridlines, adjust column widths to make all data visible .	5	
Print a copy of the worksheet with gridlines, adjust column widths to make all data visible and the formulas showing .	5	
Subtotal	180	

Job 2 (Formatting worksheet)	Points Possible	Points Earned
Member ID and Job # in right section of footer	10	
Merge and center A1 through K1	5	
Merge and center the subtitle A2 through K2	5	
Format the company name to Heading 1 cell style	5	
Format the subtitle to Heading 4 cell style	5	
Merge and right align L1:M1, L3:M3, and L5:M5	10	
Format the following cells to a percentage: C4:C12, E4:E12, G4:G12, I4:I12	5	
Format the following cells to a percentage with 1 decimal: K4:K12 and N5	5	
Format the following cells to a number with zero decimal places: B15, D15, F15, and H15.	5	
Sort the data on cities in numerical order by percentage of Total, High to Low	10	
Center align cells B3:K3	5	
Place a left border line and right border line outside of the month to group the month and percentage columns.	5	
Insert a thick bottom line for cells A3:K3 and A12:K12	5	
Fill cells J3:K12 with a light grey color	5	
Hide columns C, E, G, and I	5	
Print out the formatted spreadsheet	5	
Subtotal	95	

Job 3 (Chart)	Points Possible	Points Earned
Member ID and Job # in right section of footer	10	
Selected stacked column chart to show cities broken down by month	5	
Title of chart is Business Position Filled by City	5	
Title uses Century Gothic font, size 24, and applied an outer shadow	5	
Legend is located at the bottom of the chart	5	
Place data callout labels, customized to the percentages from the hidden columns	10	
Applied an outer shadow on the month that had the highest and the lowest for each city	5	
Only shows the horizontal labels	5	
Outlined each month's blocks with a black line.	5	
Print out the formatted chart in black and white	5	
Subtotal	60	
TOTAL POINTS	335	

JOB 1: Create Spreadsheet with Formulas and functions

1. Set the global options to include the following modifications: Verdana, 10pt, landscape, centered horizontally and vertically, fit to one page, and narrow margins.
2. Type in the following cells: A1 Digital Solution's Employment Consulting Firm, A2 Business Positions, I1 Previous Year Total, I3 This Year, I5 % Increase, J1 4277
3. Type the information in the image below:

	A	B	C	D	E	F	G
2	Business Positions						
3		January	February	March	April	Total	% of Total
4	Provo	122	101	218	178	619	
5	Boise	98	77	99	133	407	
6	Toledo	121	155	165	102	543	
7	San Francisco	131	233	301	188	853	
8	San Antonio	181	201	288	203	873	
9	Toronto	175	122	104	88	489	
10	Minneapolis	68	62	112	133	375	
11	Boston	93	102	178	151	524	
12	Charlotte	121	180	203	197	701	
13	Total						
14							
15	Average						
16	Maximum						
17	Minimum						

4. In cell F4, place a formula that calculates the total four months per city and fill down to cell F12.
5. In Cell B13 place a function that calculates each month's positions filled and fill to cell F13.
6. In Cell G4, place a formula that calculates the percentage of the total and fill to cell G12.
7. Inset a function in Cell B15 that averages January's filled positions for the month and fill to cell E15.
8. Insert a function in cell B16 that calculates the maximum filled positions for the month and fill to cell E16.
9. Insert a function in cell B17 that calculates the lowest filled positions for the month and fill to cell E17.
10. Place the cell location for the total filled positions for all cities and months in cell J3.
11. In cell J5 place a complex formula that calculates the % increase in filled positions from last year to this year.
12. Insert a column to the right of each month.
13. Title the cell of the added columns to the month % (January %).
14. Insert a formula to calculate the percentage of each month for each city with the total of that month.
15. Print a copy of the worksheet with gridlines, adjust column widths to **make all data visible**.
16. Print a copy of the worksheet with gridlines, adjust column widths to make all data visible, and the **formulas showing**.

JOB 2: Format the Spreadsheet you created in Job 1 using the following instructions:

1. Merge and center the title A1 through K1.
2. Merge and center the subtitle A2 through K2.
3. Format the company name to Heading 1 cell style.
4. Format the subtitle to Heading 4 cell style
5. Merge and right align L1:M1, L3:M3, and L5:M5
6. Format the following cells to a percentage: C4:C12, E4:E12, G4:G12, I4:I12
7. Format the following cell to a percentage with 1 decimal: K4:K12 and N5.
8. Format the following cells to a number with zero decimal places: B15, D15, F15, and H15.
9. Sort the data on cities in numerical order by percentage of Total, High to Low.
10. Center Align cells B3:K12
11. Place a left border line and right border line outside of the month to group the month and percentage columns.
12. Insert a thick bottom line for cells A3:K3 and A12:K12
13. Fill cells J3:K12 with a light grey color.
14. Hide columns C, E, G, and I
15. Print out the formatted spreadsheet.

JOB 3: Create a Custom Chart

Create a stacked column chart from the spreadsheet you have created in the previous jobs showing the number of positions for each city for each month. Format the chart to look like the following.

- Select the Stacked Column Chart to show cities broken down by month.
- Title the chart Business Positions Filled by City.
- Title is Century Gothic font size 24.
- Apply an outer shadow to the title.
- Legend located at the bottom of the chart.
- Place data callout labels, customized to the percentage from the hidden columns.
- Apply an outer shadow on the month that had the highest and lowest for each city.
- Only show the horizontal labels.
- Outline each month's block with a black line.
- Print out the formatted chart in black and white.

You should have the following four printouts:

Job 1—Unformatted

Job 1—Formulas

Job 2—Formatted

Job 3—Customized Stacked Column Chart